

Block-Type Heat Substation Sizing

Complete the online questionnaire to size an ANHEL® block-type heat substation. A manager will respond within one business day.

HOW TO COMPLETE THIS FORM

Fill in the form by hand or in a PDF reader. Tick the relevant boxes (☐ → ☒). Where parameters are unknown, leave the line blank or write «to be determined». Return the completed form to info@anhelspb.com — our engineering team replies within one business day with a sizing proposal.

* — required field.

STEP 01

Contacts and facility

Who you are and where the heat substation will operate

Contact details

Company *

Full name *

Position *

City *

Email *

Contact phone *

Facility

Facility name and location

How did you hear about us?

How did you hear about us?

- ☐ Yandex / Google ads
- ☐ Yandex / Google search
- ☐ Social networks
- ☐ Recommendation from a colleague
- ☐ Already familiar, worked with us before
- ☐ Other

Please specify

STEP 02

Inlet parameters

District-heating water parameters and heat carrier

District water temperature schedule

At the substation inlet / outlet (winter)

T1 (inlet) °C T2 (outlet) °C

District water pressure

At the substation inlet / outlet

P1 (inlet) bar P2 (outlet) bar

General

Building height m

Heat carrier

Water, glycol solution (%), etc.

STEP 03

Heating

Heating-system parameters

Heating

Thermal load Gcal/h

☐ **Dependent (direct)**
 Connection scheme

☐ **Direct**
 Connection scheme

☐ **Brazed**
 Plate heat-exchanger type

☐ **Independent (indirect)**
 Connection scheme

☐ **Gasketed (demountable)**
 Plate heat-exchanger type

☐
Shell-and-tube
 Plate heat-exchanger type

T1.2 inlet
 _____ °C

T2.2 return
 _____ °C

Heating system temperature schedule (winter)

System pressure drop
 _____ m H₂O

Maximum working pressure
 _____ m H₂O

System volume
 _____ m³

Heat-exchanger redundancy

☐ Yes
 ☐ No

HEX backup
 _____ %

☐
 100%
 Pump redundancy

☐
Spare on stock
 Pump redundancy

☐
Twin pump
 Pump redundancy

Pump VFD speed control

☐ Yes
 ☐ No

STEP 04

Ventilation

Ventilation-system parameters

Ventilation

Thermal load
 _____ kW

☐
Dependent (direct)
 Connection scheme

☐
Independent (indirect)
 Connection scheme

☐
Direct
 Connection scheme

☐
Gasketed (demountable)
 Plate heat-exchanger type

☐
Brazed
 Plate heat-exchanger type

☐
Shell-and-tube
 Plate heat-exchanger type

T1.2 inlet
 _____ °C

T2.2 return
 _____ °C

Ventilation system temperature schedule (winter)

System pressure drop
_____ m H₂O

Maximum working pressure
_____ m H₂O

System volume _____ m³

Heat-exchanger redundancy

- ☐ Yes
☐ No

HEX backup _____ % ☐ 100%
Pump redundancy

☐ Spare on stock
Pump redundancy

☐ Twin pump
Pump redundancy

Pump VFD speed control

- ☐ Yes
☐ No

STEP 05

DHW

Domestic hot water

Load and temperatures

Thermal load _____ Gcal/h

Cold water temperature _____ °C Hot water temperature _____ °C

Cold water inlet pressure at substation _____ bar Required hot water pressure _____ bar

Circulation

DHW circulation line required?

- ☐ Yes
☐ No

DHW circulation flow as % of peak demand _____ % DHW circulation head loss _____ m H₂O

DHW heat-exchanger configuration

- ☐ Single-stage ☐ Two-stage
☐ Single-skid (monoblock)

Plate heat-exchanger type

☐ Brazed

☐ Gasketed (demountable)

☐ Shell-and-tube

Redundancy

Heat-exchanger redundancy

- ☐ Yes
☐ No

HEX backup

_____ % ☐ 100%
Pump redundancy

☐ Spare on stock
Pump redundancy

☐ Twin pump
Pump redundancy

Pump VFD speed control

- ☐ Yes
☐ No

STEP 06

Additional equipment

Options, valves, communication, power

Additional equipment, functions, and parameters

Outdoor-temperature compensation

- ☐ Yes
☐ No

Automatic make-up line for heating and ventilation

- ☐ Yes
☐ No

Automatic pressure-maintenance unit for heating and ventilation

- ☐ Yes
☐ No

Thermal energy metering unit

- ☐ Yes
☐ No

Differential pressure regulator

- ☐ Yes
☐ No

Expansion tank

- ☐ Yes
☐ No

Cold-water flow meter

- ☐ Yes
☐ No

Pump-failure sensor (DP switch)

- ☐ Yes
☐ No

SCADA / dispatching

- ☐ Yes
☐ No

Provide a make-up valve

- ☐ Yes
☐ No

Provide a make-up pump

- ☐ Yes
☐ No

Pipework insulation

- ☐ Yes
☐ No

Inlet valves and fittings (steel)

- ☐ Welded
☐ Threaded
☐ Flanged

Dimensions

Substation room dimensions (L × W × H) _____ mm

Doorway size (W × H) _____ mm

Data communication to SCADA

- ☐ RS232 (485)
☐ GSM
☐ Ethernet
☐ Tel. modem

Power

Pumps powered from the substation control cabinet

- ☐ Yes
☐ No

Pumps powered from an external cabinet

- ☐ Yes
☐ No

- ☐ 1 × 230 V
Supply voltage
☐ 3 × 380 V
Supply voltage

Additional information

Additional information

Attach any project drawings, water analyses or specifications as separate files when you reply.